Solution

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ -2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}.$$

Find the following, if possible.

5 (a) A^{-1} 2 (b) B^{-1} 3 (c) C^{-1}

(a) $\left[\begin{array}{ccc|ccc} 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 3 & 0 & 3 & 1 & -1 & 0 \\ 0 & 1 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$\begin{array}{l} R_1 - 3R_3 \\ R_2 + 3R_3 \end{array} \left[\begin{array}{ccc|ccc} 3 & 0 & 0 & 1 & -1 & -3 \\ 0 & 1 & 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1/3} A^{-1} = \begin{bmatrix} 1/3 & -1/3 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

(b) Not defined (c) $\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}^{-1} = \frac{1}{7}\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$

Version 3J

(a) $\left[\begin{array}{ccc|ccc} 1 & -3 & 1 & 1 & 0 & 0 \\ 0 & 2 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 + 3/2 R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -7/2 & 1 & 3/2 & 0 \\ 0 & 2 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$\begin{array}{l} R_1 + 7/2 R_3 \\ R_2 + 3R_3 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 3/2 & 7/2 \\ 0 & 2 & 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2/2} A^{-1} = \begin{bmatrix} 1 & 3/2 & 7/2 \\ 0 & 1/2 & 3/2 \\ 0 & 0 & 1 \end{bmatrix}$

(b) Not defined

(c) $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}^{-1} = \frac{1}{-7}\begin{bmatrix} -1 & -2 \\ -3 & 1 \end{bmatrix}$

Solution

Matrix Theory, Quiz 3L — February 13, 2026

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & -2 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 3 \\ 2 & 1 \end{bmatrix}.$$

Find the following, if possible.

5 (a) A^{-1} 2 (b) B^{-1} 3 (c) C^{-1}

(a) $\left[\begin{array}{ccc|ccc} 3 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[-R_3]{R_1 - R_2} \left[\begin{array}{ccc|ccc} 3 & 0 & 4 & 1 & -1 & 0 \\ 0 & 1 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right] \xrightarrow[R_2+3R_3]{R_1-4R_3} \left[\begin{array}{ccc|ccc} 3 & 0 & 0 & 1 & -1 & +4 \\ 0 & 1 & 0 & 0 & 1 & -3 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right]$

$\xrightarrow{R_1/3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/3 & -1/3 & 4/3 \\ 0 & 1 & 0 & 0 & 1 & -3 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right]$ $A^{-1} = \begin{bmatrix} 1/3 & -1/3 & 4/3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix}$

(b) Not defined

(c) $C^{-1} = \begin{bmatrix} -1 & 3 \\ 2 & 1 \end{bmatrix}^{-1} = \frac{1}{-1-6}\begin{bmatrix} 1 & -3 \\ -2 & -1 \end{bmatrix}$

Version 3K: (a) $\left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 1 & 0 & 0 \\ 0 & 2 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 - 3/2 R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -7/2 & 1 & -3/2 & 0 \\ 0 & 2 & -3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$\xrightarrow[R_2+3R_3]{R_1-7/2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -3/2 & -7/2 \\ 0 & 2 & 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{1/2 R_2}$ $A^{-1} = \begin{bmatrix} 1 & -3/2 & -7/2 \\ 0 & 1/2 & 3/2 \\ 0 & 0 & 1 \end{bmatrix}$

(b) undefined

(c) $\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}^{-1} = \frac{1}{7}\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$